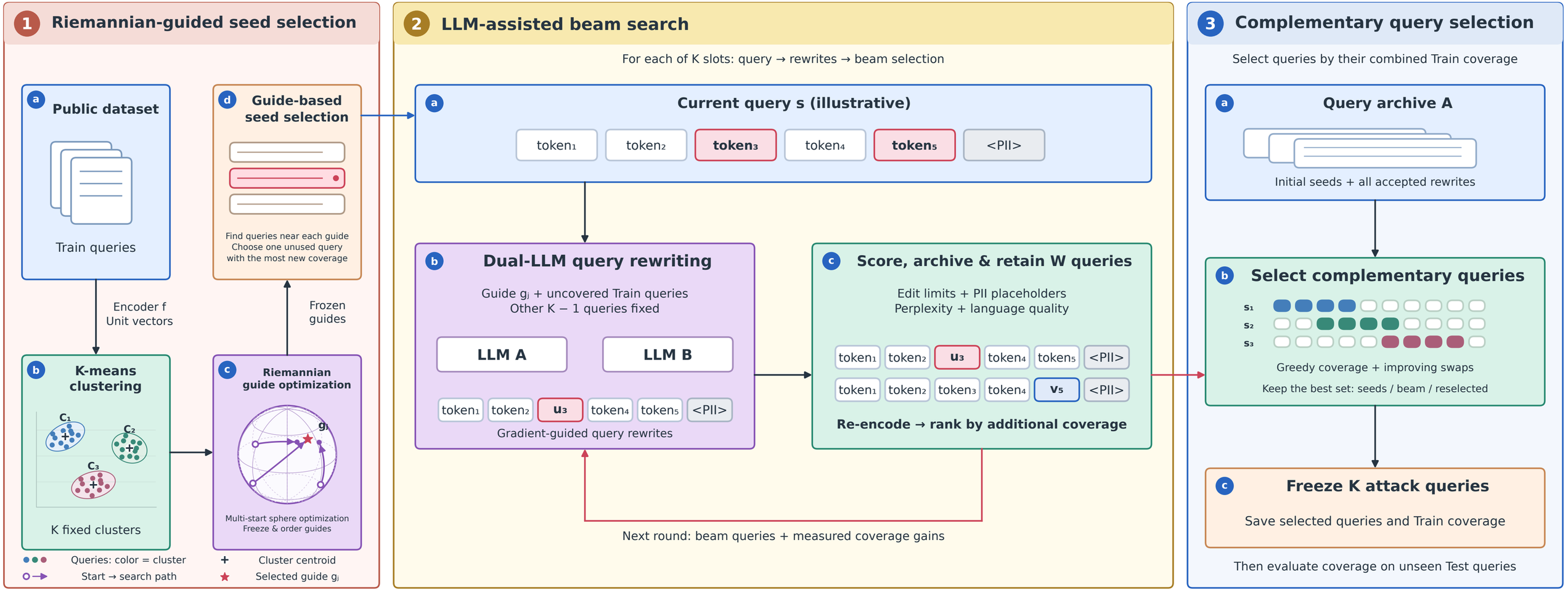

1
Riemannian-guided seed selection
a
Public dataset
Train queries
Encoder f
Unit vectors
b
K-means clustering
C1
C2
C3
K fixed clusters
d
Guide-based seed selection
Find queries near each guide
Choose one unused query
with the most new coverage
Frozen guides
c
Riemannian guide optimization
gj
Multi-start sphere optimization
Freeze & order guides
Queries: color = cluster
Start → search path
Cluster centroid
Selected guide gj
2
LLM-assisted beam search
For each of K slots: query → rewrites → beam selection
a
Current query s (illustrative)
token1
token2
token3
token4
token5
<PII>
b
Dual-LLM query rewriting
Guide gj + uncovered Train queries
Other K − 1 queries fixed
LLM A
LLM B
u3
Gradient-guided query rewrites
c
Score, archive & retain W queries
Edit limits + PII placeholders
Perplexity + language quality
v5
Re-encode → rank by additional coverage
Next round: beam queries + measured coverage gains
3
Complementary query selection
Select queries by their combined Train coverage
a
Query archive A
Initial seeds + all accepted rewrites
b
Select complementary queries
s1
s2
s3
Greedy coverage + improving swaps
Keep the best set: seeds / beam / reselected
c
Freeze K attack queries
Save selected queries and Train coverage
Then evaluate coverage on unseen Test queries